When the Supply Chain Shapes Long-Term Grid Planning

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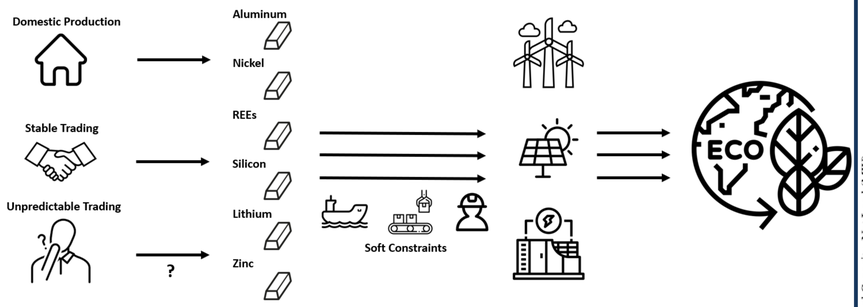
Motivation: Why Supply Chains Matter

Ensuring a reliable and resilient power grid requires accounting for supply chain constraints beyond market design, as they ultimately determine what can be built.



- Rising **demand for critical materials** → bottlenecks from concentrated supply.
- Models treat supply chains as costs, but **no materials → no deployment** possible

Research Questions:

- **What are the upstream supply chains for generation resources?**
- **How can we integrate supply chain dynamics into capacity expansion planning models?**

Contributions:

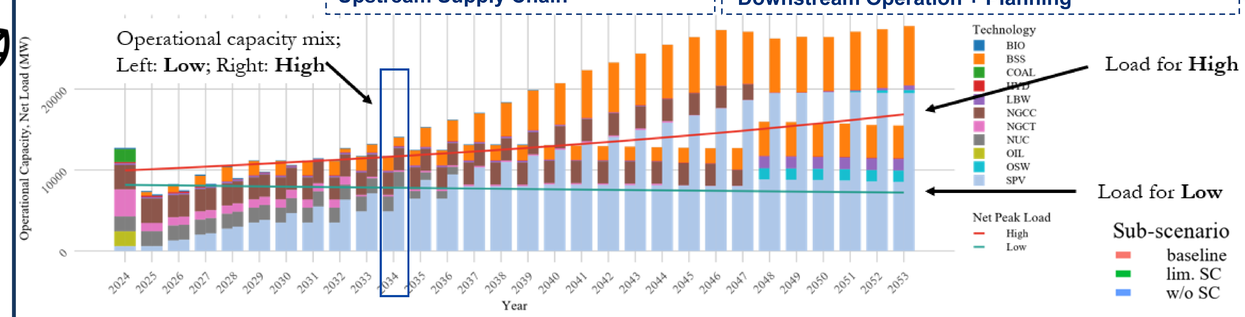
- **Supply Chain Mapping: Identification and quantification of critical materials for wind, solar PV, and battery technologies.**
- **Model Integration: Development of an optimization framework linking upstream supply chains with multi-year capacity expansion.**

Integrating Upstream Supply Chains into Capacity Expansion Planning

Objective = Investment + Operations + Penalties
Build new capacity, run the system, cover shortfalls.

Upstream Supply Chain
1. Materials → components → products.
2. Limits by production & lead times.
3. Field availability.
4. Recovery from retired units.

Downstream Operation + Planning
5. Power Balance.
6. Generation, line, storage limits.
7. Evolution: unit builds & retirements (binary).
8. Reserve margin (ELCC) + policy compliance.



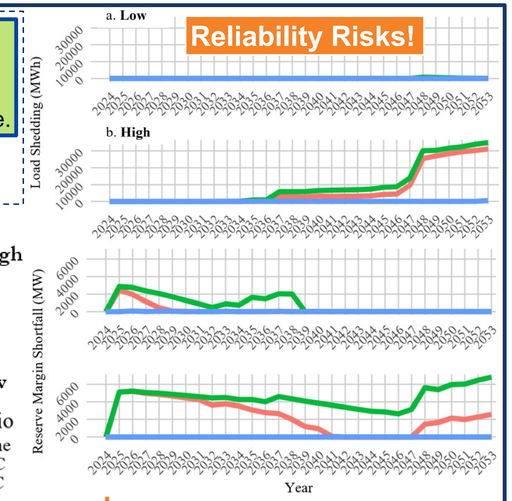
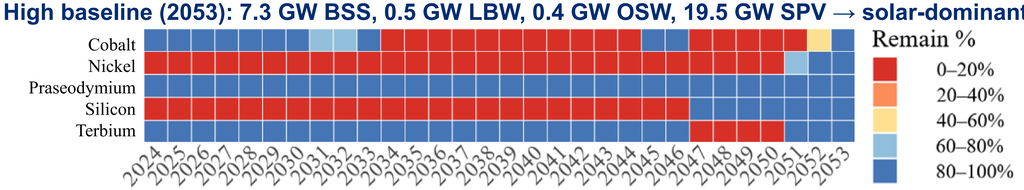


Fig. 3. Operational capacity for Low (left) and High (right) baselines. Bars = technology-specific capacity; lines = net peak load.

Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → diversified mix.
High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant.

Why this shift?



Driven by nickel and cobalt scarcity, which limits wind deployment.

Rising demand + material scarcity → **reduced flexibility and feasibility**

Note: Representative days flatten peak loads; load shedding would likely appear earlier in full chronology.

Upstream Supply Chain Bottlenecks for Wind, Solar PV, and Batteries

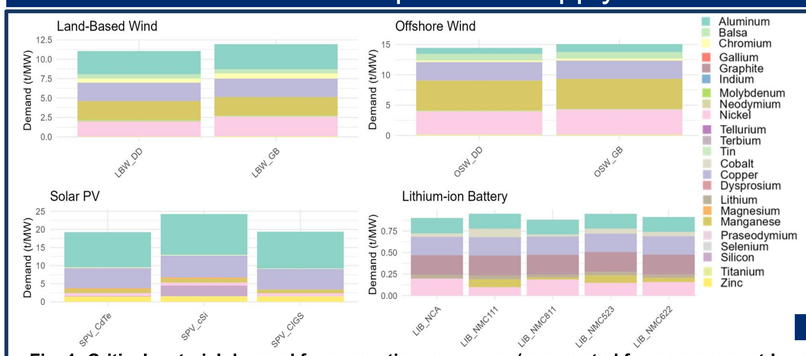


Fig. 1: Critical material demand for generation resources (aggregated from component-level)

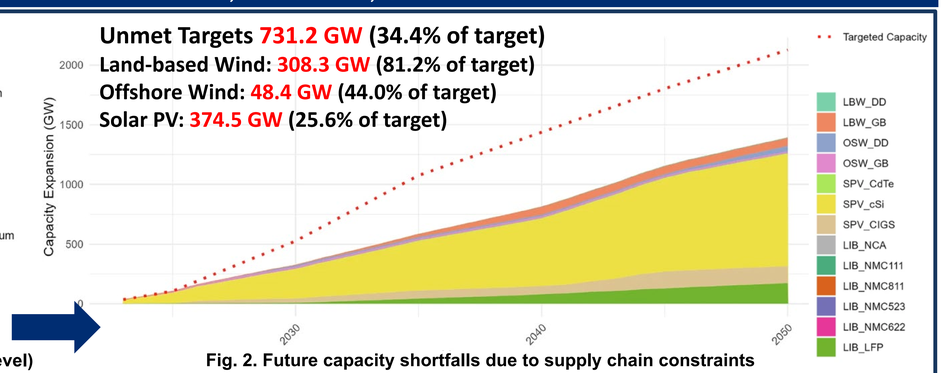


Fig. 2. Future capacity shortfalls due to supply chain constraints

- **26 critical materials identified based on the overlap of the USGS and DOE critical materials lists [1,2].**
- **Future material availability projected using the last supercycle.**
- **Optimized material use tested against 2050 net-zero goals (400 GW LBW, 110 GW OSW, 1600 GW SPV, 125 GW BSS).**

- **Net-zero capacity targets cannot be met—about 1/3 of capacity remains unmet due to material and supply chain constraints.**
- **Additional materials required, e.g., Neodymium: 7,591 metric tons (+67.5%); Nickel: 1,355,659 metric tons (+106.2%); Silicon: 913,258 metric tons (+32.1%); Tellurium: 2,834 metric tons (+327%).**

Key Takeaways

- Ignoring supply chains underestimates cost and reliability risks.
- Just-in-time planning is infeasible → requires early, strategic investments.
- No materials, no MW → diversification across technologies is essential.
- Supply chains are a critical determinant of grid feasibility and reliability.

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Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., & Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(1), 9.

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References:

[1] U.S. Geological Survey (2024). Mineral Commodity Summaries 2024. USGS, 212 p. https://doi.org/10.3133/mcs2024.
[2] Bauer, D. J., et al. (2023). 2023 Critical Materials Strategy. INL Report, Idaho National Laboratory.